

DOUBLE-ADJUGATE DYNAMICS OVER FINITE FIELDS: SINGULAR COLLAPSE AND PROJECTIVE IMAGE STAIRCASES

ANONYMOUS

ABSTRACT. For $d \geq 3$ and a finite field $\mathbb{F}_q$, we study the self-map of the full matrix space

$$\Psi(A) = \text{adj}(\text{adj } A), \quad A \in M_d(\mathbb{F}_q).$$

Jacobi's classical identity gives $\Psi(A) = \det(A)^{d-2}A$; we assign that identity zero novelty credit and determine its finite temporal consequences. Every singular matrix enters zero in one step. With $\alpha = (d-1)^2$ and $E_k = (\alpha^k - 1)/d$, every invertible matrix satisfies

$$\Psi^k(A) = \det(A)^{E_k} A.$$

Consequently

$$|\text{Fix}(\Psi^k)| = 1 + |\text{SL}_d(q)| \gcd(E_k, q-1), \quad |\text{im}(\Psi^k)| = 1 + \frac{|\text{GL}_d(q)|}{\gcd(\alpha^k, q-1)}.$$

The second formula yields the recurrent-core size and a sharp prime-valuation stabilization time. The first gives every cycle count and the Artin–Mazur zeta function by Möbius inversion, including alternating and third-period fixed-count anomalies. A second route computes two adjugates literally from minors and exhausts small matrix spaces. The note is owner-subtracted against Jacobi identities, projective adjugate/Cremona maps, and scalar power-map functional graphs; external release remains on hold.

1. INTRODUCTION

The adjugate is simultaneously a polynomial matrix operation and, on invertible matrices, determinant times inversion. Its projectivization is a classical Cremona transformation [1]. Jacobi's complementary-minor identity and its hyperadjugate extensions are established material [2]; standard finite-field facts used below can be found in [3]. Functional graphs of power maps on finite groups likewise have a direct modern literature [4]. None of those ingredients is claimed here.

We instead ask what happens when the adjugate is applied twice at every time step on the *entire* finite matrix space. Two regimes coexist. All singular ranks collapse immediately to zero, whereas every invertible projective scalar line is invariant and carries an affine power map on its nonzero scalars. This separation makes both the fixed sequence and the shrinking image sequence exact. Their arithmetic is different: fixed counts see $(\alpha^k - 1)/d$, while image sizes see α^k itself.

The residual contribution is the conjunction of the iterate normal form, singular basin, fixed and image formulas, valuation stabilization, and full cycle/zeta census over finite fields. A bounded source search did not find this temporal package, but it is not an absolute novelty certificate. External submission, public release, and priority language remain **HOLD**.

2. NORMAL FORM AND SINGULAR COLLAPSE

Fix a prime power q and an integer $d \geq 3$. Put

$$X_{q,d} = M_d(\mathbb{F}_q), \quad \Psi(A) = \text{adj}(\text{adj } A).$$

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The restriction $d \geq 3$ is structural: for $d = 2$, adjugation is already a linear involution and its square is the identity.

Lemma 2.1 (Double-adjugate identity). *For every $A \in M_d(\mathbb{F}_q)$,*

$$(1) \quad \Psi(A) = \det(A)^{d-2} A.$$

In particular, every singular matrix maps to zero, and zero is fixed.

Proof. For invertible A , the identities $\text{adj}(A) = \det(A)A^{-1}$ and $\det(\text{adj } A) = \det(A)^{d-1}$ give (1) immediately. Both sides of (1) are polynomial in the matrix entries, and Jacobi's identity extends the equality to all matrices [2]. Alternatively, the rank- $d-1$ case has $\text{rank}(\text{adj } A) = 1$, whose adjugate vanishes for $d \geq 3$; lower ranks already have $\text{adj } A = 0$. If $\det A = 0$, the right side of (1) is zero. $\square$

Define

$$(2) \quad \alpha = (d-1)^2, \quad E_k = \frac{\alpha^k - 1}{d} \quad (k \geq 1).$$

The quotient is integral because $\alpha \equiv 1 \pmod{d}$.

Theorem 2.2 (Iterate normal form). *If $A \in \text{GL}_d(q)$ and $\delta = \det A$, then for every $k \geq 1$,*

$$(3) \quad \boxed{\Psi^k(A) = \delta^{E_k} A}, \quad \det(\Psi^k(A)) = \delta^{\alpha^k}.$$

Every nonzero singular matrix has transient depth one and then stays at zero.

Proof. By lemma 2.1, if B is invertible with determinant η , then $\Psi(B) = \eta^{d-2}B$ and

$$\det \Psi(B) = \eta^{d(d-2)+1} = \eta^{(d-1)^2} = \eta^\alpha.$$

Thus the scalar exponent in the k th iterate obeys $E_{k+1} = E_k + (d-2)\alpha^k$, with $E_1 = d-2$. Since $\alpha - 1 = d(d-2)$, summing the geometric progression gives $E_k = (\alpha^k - 1)/d$. The singular assertion follows from lemma 2.1. $\square$

3. FIXED SEQUENCE AND CYCLE CENSUS

Write

$$(4) \quad |\text{GL}_d(q)| = \prod_{j=0}^{d-1} (q^d - q^j), \quad |\text{SL}_d(q)| = \frac{|\text{GL}_d(q)|}{q-1}.$$

Theorem 3.1 (All iterate-fixed counts). *For every $k \geq 1$,*

$$(5) \quad \boxed{|\text{Fix}(\Psi^k)| = 1 + |\text{SL}_d(q)| \gcd(E_k, q-1)}.$$

Consequently, the number C_r of cycles of exact length r is

$$(6) \quad C_r = \frac{1}{r} \sum_{e|r} \mu(r/e) (1 + |\text{SL}_d(q)| \gcd(E_e, q-1)),$$

and

$$(7) \quad \zeta_\Psi(z) = \prod_{r \geq 1} (1 - z^r)^{-C_r} = \exp \left(\sum_{k \geq 1} \frac{1 + |\text{SL}_d(q)| \gcd(E_k, q-1)}{k} z^k \right).$$

Proof. A singular matrix fixed by a positive iterate must equal zero. For $A \in \text{GL}_d(q)$, theorem 2.2 gives $\Psi^k(A) = A$ exactly when $(\det A)^{E_k} = 1$. The cyclic group $\mathbb{F}_q^\times$ has $\gcd(E_k, q-1)$ such determinants, and every determinant fiber in $\text{GL}_d(q)$ has cardinality $|\text{SL}_d(q)|$. This proves (5). Möbius inversion of iterate-fixed counts gives (6), and the standard finite-map cycle factorization gives (7). $\square$

The formula can show large low-period jumps even though projective matrix classes never move. Two registered examples are

$$(q, d) = (5, 4) : |\text{Fix}(\Psi^k)| = 58\,032\,000\,001, 116\,064\,000\,001, \dots,$$

$$(q, d) = (7, 3) : |\text{Fix}(\Psi^k)| = 5\,630\,689, 5\,630\,689, 16\,892\,065, \dots$$

The first sequence alternates; the second has a jump precisely at multiples of three. These are direct gcd effects in (5), not numerical experiments.

4. PROJECTIVE IMAGE STAIRCASE AND RECURRENT CORE

Let $[A] = \{cA : c \in \mathbb{F}_q^\times\}$ be the projective scalar line through an invertible matrix. Every such line has $q - 1$ points, and distinct lines are disjoint.

Theorem 4.1 (Exact images). *For every $k \geq 1$,*

$$(8) \quad |\text{im}(\Psi^k)| = 1 + \frac{|\text{GL}_d(q)|}{\gcd(\alpha^k, q - 1)}.$$

Let

$$(9) \quad b = \prod_{\ell|\alpha} \ell^{v_\ell(q-1)},$$

where the product is over primes dividing α . Then

$$(10) \quad |\text{Rec}(\Psi)| = 1 + \frac{|\text{GL}_d(q)|}{b}.$$

With the empty maximum interpreted as zero, put

$$(11) \quad t_* = \max_{\ell|\gcd(\alpha, q-1)} \left\lceil \frac{v_\ell(q-1)}{v_\ell(\alpha)} \right\rceil.$$

For $k \geq 0$, write $I_k = \Psi^k(\text{GL}_d(q))$, so that $I_0 = \text{GL}_d(q)$. The nested invertible image chain $I_0 \supseteq I_1 \supseteq \dots$ stabilizes for the first time at t_* , and the maximum transient depth on all of $M_d(\mathbb{F}_q)$ is

$$(12) \quad \max\{1, t_*\}.$$

Proof. Fix $A \in \text{GL}_d(q)$ with determinant δ . For $c \in \mathbb{F}_q^\times$, [theorem 2.2](#) gives

$$(13) \quad \Psi^k(cA) = \delta^{E_k} c^{dE_k+1} A = \delta^{E_k} c^{\alpha^k} A.$$

Thus the image on each projective line is a coset of the subgroup of α^k th powers. It has $(q - 1)/\gcd(\alpha^k, q - 1)$ points. There are $|\text{GL}_d(q)|/(q - 1)$ invertible projective lines, and every singular matrix contributes only the common image zero. This proves (8).

For each prime ℓ , the exponent of ℓ in $\gcd(\alpha^k, q - 1)$ is $\min\{kv_\ell(\alpha), v_\ell(q - 1)\}$. Hence these gcDs stabilize at b , first at the time (11). In a finite functional graph the stable image is exactly the recurrent set, proving (10). Because images of a finite self-map are nested, strict image loss through time $t_* - 1$ and equality from time t_* imply that the maximum invertible tail depth is exactly t_* . This includes $t_* = 0$, when the restriction to the invertible set is a permutation. Singular points contribute depth one, proving (12). $\square$

In discrete-log coordinates on a projective line, (13) is an affine power map $x \mapsto \alpha x + h$ modulo $q - 1$. This is the second proof route for the valuation staircase: an affine translation changes cycle placement but not the image index of multiplication by α^k .

5. EXACT CONTROLS AND OWNER BOUNDARY

The accompanying program computes determinants by modular elimination and adjugates literally from signed minors. It exhausts $M_3(\mathbb{F}_2)$, $M_3(\mathbb{F}_3)$, and $M_4(\mathbb{F}_2)$, comparing two literal adjugations with (1). It then checks singular collapse, six fixed counts, and six image sizes. Separate determinant-representative lanes test twelve iterates for $(q, d) = (5, 4), (7, 3), (11, 4), (13, 5)$. Independent scalar-line lanes at $(q, d) = (5, 4), (7, 3), (17, 3), (257, 3), (19, 4)$ exercise image stabilization times $t_* = 0, 1, 2, 4, 1$, respectively, including literal strict losses and the first stable image. All arithmetic is exact. The computations are finite falsifiers, not a replacement for the proofs.

Jacobi's identity, the projective adjugate map, cyclicity of $\mathbb{F}_q^\times$, and general power-map functional graphs are established background. We claim none of them separately. The residual claim is the full-matrix temporal package (5)–(12), under a bounded collision search and an external-release hold.

Two internal boundaries are explicit. P99 obtains valuation staircases from a unipotent permutation of fixed-index sublattices; it has no singular basin, adjugate identity, or shrinking projective lines. P97 studies sumset squaring on subsets of a prime cyclic group; here the power map occurs only on scalar coordinates inside an invariant matrix line. Neither paper owns the full-matrix fixed/image pair proved here, while the shared valuation and power-map primitives receive no separate credit.

6. CONCLUSION

Double adjugation separates the finite matrix space into a singular basin that dies immediately and invertible projective lines carrying affine power maps. This yields two complementary exact time series: the fixed sequence detects $E_k = (\alpha^k - 1)/d$, while the image staircase detects the prime valuations of α^k against $q - 1$. Together they determine every cycle count, the zeta function, the recurrent core, and the sharp transient depth.

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